



Steward Systems Behavioural Intelligence Agent

A Nigerian-Contextualised LLM Agent for Review Simulation and Personalised Recommendation

Solution Paper for the DSN x BCT LLM Agent Challenge / Data & AI Summit Hackathon 3.0

Team	Steward Systems
Live Agent	https://steward-systems-bct-llm-agent-challenge.onrender.com
Repository	https://github.com/simeonsaaaondonyam100-hub/steward-systems-bct-llm-agent-challenge

Abstract

This paper presents the Steward Systems Behavioural Intelligence Agent, a containerised FastAPI system that solves the two required challenge tasks: user modelling/review simulation and personalised recommendation. The system builds behavioural profiles from persona text and past reviews, predicts star ratings, generates human-readable reviews, and ranks candidate items with transparent scoring. The agent is intentionally Nigerian-contextualised: it models value-for-money, delivery delay, pepper/spice preference, portion size, local food cues, and practical consumer expectations. The default submission is lightweight and reproducible with bundled sample data, while the repository also includes optional Yelp Open Dataset ingestion for real-world user-item-review experimentation.

1. Problem Formulation

Task A requires an agent that receives a user persona, review history, and target item details, then returns a predicted rating and a plausible written review. Task B requires an agent that receives a user persona and context, then returns ranked recommendations tailored to the user. The central design choice was to treat both tasks as two specialised capabilities of one behavioural intelligence system, not as two disconnected scripts.

The finished application exposes a live landing page, a health endpoint, and Swagger/OpenAPI documentation. The live agent is available at <https://steward-systems-bct-llm-agent-challenge.onrender.com>, while the reproducible codebase is available through GitHub and Docker.

2. Data and Preprocessing

The default submitted demo uses a lightweight bundled sample dataset with 18 users, 60 items, and 160 reviews across food, drinks, books, movies/entertainment, and everyday products. This keeps the container fast, deterministic, and judge-friendly while still demonstrating behavioural variation, cold-start handling, cross-domain recommendation, rating strictness, and Nigerian consumer context.

The repository also includes optional Yelp Open Dataset ingestion support. The full Yelp dataset is not bundled because of size, but the ingestion script maps Yelp business, review, and user JSON files into the same internal user-item-review schema used by the agent. This preserves compatibility with real-world review datasets without forcing judges to download large raw files before testing the default demo.

The internal schema separates users, items, and reviews. Review records provide the behavioural evidence for rating tendencies, preference cues, complaints, category affinity, and tone. Item records provide metadata such as category, tags, location, price, delivery timing, portion size, popularity score, and average rating where available.

3. System Architecture

The project is implemented as a modular FastAPI application. Thin API routes delegate work to agent and service layers. The Review Simulation Agent handles Task A; the Recommendation Agent handles Task

B. Both share user profiling, Nigerian-context interpretation, semantic similarity fallback, data loading, and evaluation utilities. The application is Dockerised and deployment-safe: the container binds to 0.0.0.0 and uses the platform-provided PORT variable when available, defaulting to 8000 locally.

The live deployment is hosted on Render. The root route provides a lightweight HTML landing page, /health provides a service-health check, and /docs exposes Swagger/OpenAPI for interactive testing. The API documentation contains realistic request examples for both tasks, and additional copyable payloads are documented in docs/sample_payloads.md. The app requires no database, Supabase backend, paid API key, or external model service for the default demo.

3.1 API Guardrails and Input Validation

A final guardrail layer prevents meaningless Swagger placeholders such as "string", 0, empty strings, null, and n/a from being interpreted as user preferences or item signals. Task A rejects placeholder-only item names and categories with validation errors. Task B treats placeholder-only or insufficient profile evidence as cold-start input, rather than pretending that personal history exists.

The response language was also polished so that generated reviews sound human-facing, while technical reasoning remains in structured explanation fields. This protects the demo from accidental placeholder submissions and improves reliability during manual judging.

4. Task A Method: Review Simulation

Task A proceeds in four stages. First, the agent builds a behavioural profile from the persona and past reviews, estimating rating strictness, preferred qualities, common complaints, tone, price sensitivity, spice preference, portion preference, and category affinity. Second, it compares the target item to this profile using item metadata and text cues. Third, it predicts a rating bounded between 1 and 5 and returns an integer `predicted_star_rating` for judge-friendly interpretation. Finally, it generates a written review that mentions the target item, reflects the predicted rating, and uses natural Nigerian-context-aware language without overusing slang.

For example, a Lagos-based student who likes spicy, affordable meals and quick delivery receives a high rating for Spicy Chicken Shawarma when the item matches pepper preference, reasonable delivery time, and value expectations. Positive and negative signals remain visible for explainability, but the generated review itself avoids internal modelling phrases.

5. Task B Method: Personalised Recommendation

Task B builds a user profile, loads candidate items, computes hybrid scores, and returns ranked recommendations. The scoring formula combines semantic similarity, preference match, context match, item quality/popularity, Nigerian-context match, and penalties for disliked traits. The selected weights are: semantic 0.18, preference 0.38, context 0.20, quality 0.10, Nigerian context 0.10, and penalty 0.04.

Cold-start handling is explicit. If user evidence is missing or placeholder-only, the agent ignores non-informative fields and falls back to broad popularity, item quality, and Nigerian-context fit. If real history exists, past reviews influence profile strictness, preferences, complaints, and ranking explanations. The returned response preserves `score_breakdown`, `context_fit`, `preference_match_explanation`, `ranking_formula`, and `semantic_mode` for transparency.

6. Nigerian Contextualisation Layer

The NigerianContextAdapter makes the system locally intelligent rather than merely generic. It accounts for affordability, value for money, portion size, pepper/spice preference, delivery timing, Lagos-style order expectations, and familiar local food/drink cues such as jollof rice, suya, shawarma, pepper soup, ofada, bole, zobo, Chapman, small chops, and similar examples. The aim is not gimmicky slang, but practical cultural relevance.

This layer improves review generation and recommendation explanations. A budget-conscious user can receive reasons such as “price looks reasonable for a budget-conscious Nigerian user,” while a dinner-context recommendation can prioritise large portions and filling meals. The ablation results show that the Nigerian-context layer improves Hit Rate@5 and contextual fit in the bundled evaluation.

7. Evaluation and Results

Evaluation is designed for transparency rather than inflated claims. The included dataset is lightweight and reproducible, so the metrics demonstrate the behaviour of the submitted system rather than claiming full-scale industrial performance. Task A uses RMSE and MAE for rating prediction. Task B uses Hit Rate@K and NDCG@K, with ablation comparisons against popularity and content-only baselines.

Area	Metric	Result	Interpretation
Task A	RMSE	0.703	Rating prediction error on the reproducible sample set.
Task A	MAE	0.546	Average absolute rating error.
Task B	Hit Rate@5	0.429	Relevant items often surface within top 5.
Task B	Hit Rate@10	0.571	Relevant items appear in top 10 for most test cases.
Task B	NDCG@10	0.207	Ranking order is useful but can improve with larger real-world data.
Tests	Pytest	18 passed	Core endpoints, guardrails, ingestion and response quality covered.

The final manual API tests were also successful. Task A returned a natural review and predicted_star_rating for a realistic shawarma payload. Task B returned food recommendations such as Bole and Fish Combo, Jollof Rice and Grilled Chicken, Ofada Rice and Ayamase, Beans and Plantain Bowl, and Spicy Chicken Shawarma for a Lagos student dinner context.

7.1 Ablation Summary

Configuration	HR@5	HR@10	NDCG@5	NDCG@10
Popularity baseline	0.286	0.429	0.143	0.169
Content-only baseline	0.000	0.286	0.000	0.048

Hybrid without Nigerian context	0.286	0.571	0.121	0.194
Hybrid with Nigerian context	0.429	0.571	0.155	0.207

The hybrid approach outperforms the popularity baseline on Hit Rate@5 and matches or improves Hit Rate@10. Nigerian context provides its clearest benefit in top-5 relevance and human-readable explanation quality. Ranking order can improve further with larger real-world datasets and learned embeddings, which are documented as future work.

8. Reproducibility, Deployment, and Judge Experience

The codebase is documented, modular, and reproducible. Judges can run the app locally with Docker, visit the landing page, open /docs, and execute Task A or Task B using realistic examples. The README warns that Swagger schema placeholders are not meaningful inputs and points judges to realistic Swagger examples and docs/sample_payloads.md. The Docker build has been verified locally, and the app is deployed live on Render for immediate access.

- Live agent: <https://steward-systems-bct-llm-agent-challenge.onrender.com>
- API docs: /docs
- Health check: /health
- Default mode: bundled sample data, no secrets, no database, no paid API key
- Optional mode: Yelp Open Dataset ingestion into the same internal schema

9. Limitations and Future Work

The submitted dataset is intentionally lightweight to keep the demo fast and reproducible. The system does not claim to have been trained on the full Yelp, Amazon Reviews, or Goodreads datasets. Instead, it demonstrates a complete architecture and provides an optional real-data ingestion path. The ranking metrics are useful but not final; stronger NDCG results would require larger held-out data, richer item metadata, and more systematic weight tuning.

Future work includes enabling sentence-transformer embeddings by default where model downloads are available, ingesting larger Yelp/Amazon/Goodreads subsets, improving learned ranking, expanding human evaluation, and adding a simple demo frontend if needed. The core architecture intentionally avoids unnecessary infrastructure and remains ready for real datasets without requiring Supabase or another backend database.

10. Disclosure of Frameworks, Models, and Datasets

This project uses open-source software frameworks including Python, FastAPI, Uvicorn, Docker, pytest, and standard Python data-processing and evaluation libraries. Semantic matching runs in deterministic fallback mode by default, with optional support for public pre-trained sentence-transformer embeddings where available. The submitted default demo does not require a paid API key or proprietary model service.

The submitted container includes a lightweight reproducible sample dataset. Optional Yelp Open Dataset ingestion support is included, but the full Yelp dataset is not bundled because of size. Any future use of Yelp, Amazon Reviews, Goodreads, or other external datasets should be disclosed with the dataset source, preprocessing steps, and licensing/usage constraints.

11. Conclusion

The Steward Systems Behavioural Intelligence Agent provides a complete, reproducible, Nigerian-contextualised solution for both tasks in the DSN x BCT LLM Agent Challenge. It goes beyond a prompt-only wrapper by combining behavioural profiling, guardrailed APIs, hybrid recommendation, explainable scoring, Nigerian-context reasoning, evaluation scripts, optional Yelp ingestion, Docker packaging, and a live deployment. The result is a focused agentic system that is lightweight enough for judges to run, yet structured enough to scale to real review datasets and more advanced modelling.